

Simple Unified Model

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Abstract

A Simple Approach to the Valuation of Risky Streams

(Ross 1978) showed how to value uncertain future cash flows. His two main results are:

1. Valuation is a purely geometric result that does not involve probability.
2. The theory applies to *any* portfolio of instruments.

Perhaps this was too astonishing for people to appreciate at the time. There is no need for Ito processes or partial differential equations, much less an otiose “real-world” measure that gets immediately thrown out for a “risk-neutral” measure. It also applies to *any* portfolio of instruments, not just the bond, stock, and option (Black and Scholes 1973) and (Merton 1973) considered.

Ross’s definition of a cash flow was a jump in stock price. It is well-known in the equity world that stock price jumps down by the dividend amount since you will receive that until the stock is sold after the ex-dividend date. Stock prices also jump between close and open but there is no dividend payment involved. Adding an explicit knob for cash flows to Ross’s theory leads to a simple model that adheres more closely to market realities.

If you are a familiar with the classical theory see the Appendix to scrub your brain of unnecessary mathematical accoutrements.

Market

Every *instrument* has bounded *price* X_t and *cash flow* C_t at trading time t . The cash flow is zero except at a finite number of times. If I is the collection of all instruments then we can model them simultaneously if we let X_t and C_t be vectors indexed by I .

Trading

A *trading strategy* is a finite number of increasing stopping times $\tau_0 < \dots < \tau_n$ and trades Γ_j indexed by instruments that depend only on available information. Trades accumulate to *positions*

$$\Delta_t = \sum_{\tau_j < t} \Gamma_j = \sum_{s < t} \Gamma_s$$

where $\Gamma_s = \Gamma_j$ when $s = \tau_j$ and is zero otherwise. Note the strict inequality – trades take time to settle into a position. A trading strategy is *closed-out* if $\sum_j \Gamma_j = 0$.

Accounting

Trading involves accounting. The *value* (or mark-to-market) at time t

$$V_t = (\Delta_t + \Gamma_t) \cdot X.$$

is the putative value of liquidating the existing position and trades just executed assuming that can be done at current market prices. The trades just executed are not yet in the position, but they soon will be, $\Delta_t + \Gamma_t = \Delta_{t+\epsilon}$ for some sufficiently small $\epsilon > 0$.

The *amounts* involved in trading are

$$A_t = \Delta_t \cdot C_t - \Gamma_t \cdot X_t.$$

Cash flows proportional to the existing position are credited to the trading account and trades just executed are debited at the current market prices.

Arbitrage

If there exists a trading strategy that makes money on the first trade ($A_{\tau_0} > 0$) and never loses money until the trading strategy is closed out ($A_t \geq 0, t > \tau_0$) then the model admits arbitrage. The closed out condition is necessary to prevent strategies that simply borrow money and never pay it back. Note this definition does not involve probability. Ross showed a model has no arbitrage if and only if there exist positive adapted finitely-additive measures

D_t on $\mathcal{A}_t^1$ with

$$X_t D_t = (X_u D_u + \sum_{t < s \leq u} C_s D_s)|_{\mathcal{A}_t} \quad (1)$$

where $|_{\mathcal{A}}$ indicates restriction. We can, and do, assume $D_0(\Omega) = 1$ since we can divide both sides of equation (1) by any positive constant.

We call such measures *deflators*. If repurchase agreements are available in the market then, as we will see later, the usual stochastic discount is a canonical deflator.

Ross used the Hahn-Banach theorem to prove existence but there is no need for that. Every arbitrage-free model has a simple parameterization.

Parameterization

Every arbitrage-free model is parameterized by measures indexed by instruments, $(M_t)_{t \in T}$, satisfying $M_t = M_u|_{\mathcal{A}_t}, t \leq u$ (a *martingale measure*) where

$$X_t D_t = X_0 M_t - \sum_{s \leq t} C_s D_s. \quad (2)$$

Exercise. Show equation (2) implies equation (1).

Hint: Replace t by u in equation (2) and plug that into the right-hand side of equation (1).

The Simple Unified Model only provides an elementary but mathematically rigorous framework to approach the highly non-trivial and computationally intensive problem of finding martingale measures that can fit market data.

Black-Scholes/Merton

For example, the Black-Scholes/Merton model for the bond and stock with no dividends is parameterized by $M_t = (1, e^{\sigma B_t - \sigma^2 t/2})P$, $D_t = e^{-\rho t}P$ where P is Wiener measure on $\Omega = C([0, \infty))$, the space of continuous functions from $[0, \infty)$ to the real numbers, and B_t is standard Brownian motion.

¹A partition of a set Ω is a collection of pairwise disjoint subsets of Ω having union Ω . No information corresponds to the singleton partition $\{\Omega\}$ – we only know the outcome ω is a member of Ω . Complete information corresponds to the partition of singletons $\{\{\omega\} \mid \omega \in \Omega\}$ – we know exactly which outcome occurred. Partial information corresponds to knowing only which atom of the partition the outcome belongs to.

Exercise. Show $X_t = (R_t, S_t)$ where $R_t = re^{pt}$ and $S_t = se^{\sigma B_t + (\rho - \sigma^2/2)t}$.

Hint: Use equation (2) with cash flows $C_t = (0, 0)$.

There is no need to introduce a so-called “real world” measure that requires a Nobel Prize winning argument involving partial differential equations to show it can be tossed out.

Dividends

A *fixed dividend* on a stock is a cash flow d_t at dividend date t . A *proportional dividend* p_t is a cash flow $p_t S_t$ at dividend date t . Companies often announce projected dividends several months in advance. Future dividends after that are usually assumed to be proportional to the stock price. Specifying a function C that is one at zero and decreases to zero at infinity can be used to blend these using cash flows $C(t)d_t + (1 - C(t))p_t S_t$.

Ross made the category error of defining dividends from a stock as a jump in their price. Stock prices jump for reason other than dividend payments.

Note that stock price jumping down by dividend value at dividend dates is not a consequence of a no-arbitrage strategy. It follows directly from the parameterization of equation (2).

Considering dividends expands the original sample space of all possible stock paths to contain all finite sequences $(t_j, C_{t_j}j)$ with increasing dividend dates t_j and dividend payments C_{t_j} known at time t_j . Explicitly specifying this allows us to consider, e.g., the sensitivity to dividend times and payments.

Market Parameterization

(Breedon and Litzenberger 1978) showed the risk-neutral distribution of an underlying at expiration is determined by the second derivative of European call prices with respect to strike. In practice, out-of-the-money puts are converted to calls using put-call parity and option values are interpolated between traded strikes. This market data can be used to define a martingale measure for equation (2).

If the cash flows are zero we can take $M_t = S_t D_t / S_0$ where S_t is the Breedon-Litzenberger distribution derived from option prices expiring at time t and S_0 is the current underlying price.

Exercise. If M_t is a measure on $\mathcal{A}_t$ then $M_s = M_t|_{\mathcal{A}_s}$, $s \leq t$, is a martingale measure.

Hint: For any function $f: X \rightarrow Y$ and $A \subseteq B \subseteq X$ then $f|_A = (f|_B)|_A$.

This replaces the use of conditional expectation used in classical models.

Aside from interpolation methods introducing mathematical artifacts into the option prices, the most most difficult and important problem is coming up with the sample space Ω and partitions $\mathcal{A}_t$ representing information at time t .

People tend to underestimate what should be included in a sample space and not have sufficient imagination when it comes to inventing partitions sufficiently flexible to capture possible market dynamics. The SUM only provides a framework to reason rigorously about this and a mathematical model that is straightforward to implement.

Consequences

Equation (1) and the definition of V_t and A_t result in

$$V_t D_t = (V_u D_u + \sum_{t < s \leq u} A_s D_s)|_{\mathcal{A}_t}. \quad (3)$$

The proof starts with $V_t D_t = (\Delta_t + \Gamma_t) \cdot X_t D_t$ then uses induction on $X_t D_t = (X_u D_u + C_u D_u)|_{\mathcal{A}_t}$ for some $u > t$ sufficiently small.

Note how value and amount in equation (3) correspond to price and cash flow in equation (1)

Trading strategies create synthetic market instruments.

This is the skeleton key to valuing, hedging, and managing the risk of derivative instruments.

Derivatives

A (cash settled) derivative is a contract where the seller will pay the buyer cash flows $\hat{A}_k$ at times $\hat{\tau}_k$. The buyer is willing to pay the seller to take this risk off their hands. This short note provides nothing to suggest a solution to the conundrum of what premium, vigorish, or baksheesh should be charged for that service.

A perfect hedge is a closed-out trading strategy (τ_j, Γ_j) with $A_{\hat{\tau}_k} = \hat{A}_k$ and $A_t = 0$ otherwise. The latter condition is referred to as *self-financing*. This can be achieved by investing non-zero amounts in the money-market/funding account if that is available. As (Harrison and Kreps 1979) showed, continuous

time trading is a mathematical pathology of the theory of Ito processes that leads to contradictions.

In reality, only a finite number of trades are possible and perfect hedges do not exist.

One approach to finding a hedging strategy is to note $V_t = (\Delta_t + \Gamma_t) \cdot X_t$ and that $V_t D_t = (\sum_{\hat{\tau}_k > t} \hat{A}_k D_{\hat{\tau}_k})|_{\mathcal{A}_t}$ can be calculated using the contract terms $\hat{\tau}_k$ and $\hat{A}_k$ by equation (3). The Fréchet derivative² of value V_t with respect to the underlying X_t is $D_{X_t} V_t = \Delta_t + \Gamma_t$. At time 0 the position is 0 so this gives us the initial trade Γ_0 . At any time after that we have $\Gamma_t = D_{X_t} V_t - \Delta_t$. The position Δ_t is known at time t so this can be used to determine potential trading strategies. Note Δ is delta and Γ is gamma.

What this theory does not tell us is when to trade. We could let $\tau_j = j\Delta t$ for some time increment Δt and this results in the usual Black-Scholes/Merton delta hedge as the time increment goes to zero. A smarter idea might be to set a price increment ΔX and determine the next trading time τ_{j+1} from the last trading time τ_j by $\tau_{j+1} = \inf\{t > \tau_j \mid \|X_t - X_{\tau_j}\|_\infty > \Delta X\}$.

What this theory does allow us to do is rigorously analyze any trading strategy. For instance, the above ΔX strategy could be implemented using limit orders. At time τ_j place limit orders at $X_{\tau_j} \pm \Delta X$ of size Γ_{j+1} . We do not know Γ_{j+1} at time τ_j since it depends on τ_{j+1} and $X_{\tau_{j+1}}$, however limit orders cost nothing to place and cancel so we can keep adjusting their size as $t \rightarrow \tau_{j+1}$.

Application

This sections shows how to use the SUM to value various instruments from their cash flows. The absence of arbitrage restricts price dynamics.

Zero Coupon Bond

A zero coupon bond $D(u)$ pays 1 unit at maturity u . Its price $X_t^{D(u)} = D_t(u)$ at time $t \leq u$ is determined by $D_t(u)D_t = D_u|_{\mathcal{A}_t}$ so the price $D_t(u): \mathcal{A}_t \rightarrow \mathbf{R}$

²If $F: X \rightarrow Y$ is a function between normed linear spaces the Fréchet derivative (when it exists) is the best linear approximation of F in a neighborhood of a point

$$F(x+h) = F(x) + DF(x)h + o(h), x, h \in X$$

where $DF(x)$ is a bounded linear operator from X to Y .

is $D_t(u)(A) = D_u(A)/D_t(A)$ for $A \in \mathcal{A}_t$. Deflators determines the dynamics of zero coupon bond prices.

Note $D_0(u)D_0 = D_u|_{\mathcal{A}_0}$ so $D_0(u) = D_u(\Omega)/D_0(\Omega) = D_u(\Omega)$ since we assume $D_0(\Omega) = 1$. We write³ $D(u) = D_0(u)$ for the current discount.

The *spot rate* $r(u)$ is determined by $D(u) = e^{-ur(u)}$ and the *forward rate* $f(u)$ by $D(u) = e^{-\int_0^u f(t) dt}$. Note the spot rate $r(u) = (1/u) \int_0^u f(t) dt$ is the average of the forward rate. The forward rate $f(u) = r(u) + tr'(u)$ involves the derivative of the spot rate.

For computer implementations it is preferred to use the forward rate to define the discount and spot rate since averaging smooths the data, whereas the derivative of a nearly constant spot rate can be arbitrarily large.

Canonical Deflator

Repurchase agreements determine a *canonical deflator*. A repurchase agreement over the interval from t to $t + \Delta t$ is specified by a rate f_t known at time t . The price at t is 1 and has a cash flow of $e^{f_t \Delta t} \approx 1 + f_t \Delta t$ at time $t + \Delta t$.

By equation (1) we have $D_t = e^{f_t \Delta t} D_{t+\Delta t}|_{\mathcal{A}_t}$. The canonical deflator specifies $D_{t+\Delta t}$ is known at time t so $D_t = e^{-f_t \Delta t} D_{t+\Delta t}$. Given back-to-back repos at times t_j having rates f_j we have $D_j = e^{-f_j \Delta t_j} D_{j+1}$ and $D_n = e^{-\sum_{j < n} f_j \Delta t_j} D_0$ is the canonical deflator at time t_n .

The continuous time analog is $D_u = e^{-\int_0^u f_t dt} D_0$ where f_t is the continuously compounded instantaneous forward rate at time t . This is commonly referred to as the *stochastic discount* and the reciprocal $R_t = 1/D_t$ is called a *money market account*. It is convenient to assume there is an instrument with price R_t having no cash flows in order to finance trading strategies.

It is rare for market participants to directly use repos to finance their trading. Traders working for large banks have access to a funding desk that charges them the repo rate plus a few basis points. Day traders using a credit card to fund trading get charged at the APR they qualify for. The value of an instrument depends on the credit quality of the hedger.

The SUM model does not assume a risk-free rate exists. If a funding instrument is being used then it, or all the instruments used to mimic it, should be added to the set of instruments I .

³This uses the helpfully confusing notation $D(u)$ for both the zero coupon instrument and its discount.

Forward Rate Agreement

A forward rate agreement pays -1 at *effective date* u and $1 + f\delta(u, v)$ at *termination date* $v > u$ where f is the forward rate agreement *coupon*, δ is the *day count basis*, and $\delta(u, v)$ is the *day count fraction* approximately equal to the time in years from u to v . For example, the Actual/360 day count is the number of days from u to v divided by 360. In practice, the payment dates are adjusted by holidays and business rolling conventions.

The *par forward coupon* $F^\delta(u, v)$ is the coupon making the initial price equal to zero. Since $0 = (-D_u + (1 + F^\delta(u, v))D_v)|_{\mathcal{A}_0}$ we have $F^\delta(u, v) = (D(v)/D(u) - 1)/\delta(u, v)$.

Exercise. Show the *par coupon* at time $t \leq u$ is $F_t^\delta(u, v) = (D_t(v)/D_t(u) - 1)/\delta(u, v)$.

A repo is equivalent to a forward rate agreement having day count fraction approximately equal to one day.

Fixed Income

A fixed income bond is just a finite portfolio of zero coupon bonds with cash flow c_j at times u_j . Its *present value* at time t is $P_t = \sum_{u_j > t} c_j D_t(u_j)$. The *continuously compounded yield* y_t at time t is defined by $P_t = \sum_{u_j > t} c_j e^{-(u_j - t)y_t}$. The *duration* of a bond is the first derivative with respect to yield dP_t/dy_t and the *convexity* is the second derivative d^2P_t/dy_t^2 .

Futures

A futures contract on underlying S is specified by observation dates $t_0 < \dots < t_n$. Its price is always zero and its *quote* at *expiration* t_n is $\Phi_{t_n} = S_{t_n}$. Prior to expiration the quotes Φ_t , $t < t_n$, are determined by the market. Futures pay cash flows $C_{t_j} = \Phi_{t_j} - \Phi_{t_{j-1}}$ at t_j . By equation (1), $0D_{t_{j-1}} = (\Phi_{t_j} - \Phi_{t_{j-1}})D_{t_j}$ in any arbitrage-free model.

If D_t is a canonical deflator then $\Phi_{t_{j-1}} = \Phi_{t_j}|_{\mathcal{A}_{t_{j-1}}}$ so the futures quotes are a martingale.

Cost of Carry

A forward contract with strike k expiring at u on underlying S pays exactly one cash flow $S_u - k$ at u . Its price F_t at time t satisfies $F_t D_t = ((S_u - k)D_u)|_{\mathcal{A}_t}$. In an arbitrage-free model the price of the

underlying satisfies $S_t D_t = (S_u D_u)|_{\mathcal{A}_t}$ assuming it pays no cash flows. Hence $F_t D_t = S_t D_t - k D_t(u) D_t$ so $F_t = S_t - k D_t(u)$.

The value of k making $F_t = 0$ is called the at-the-money, or par, forward and is denoted $f_t(u)$. This implies the *cost-of-carry formula* $S_t = f_t(u) D_t(u)$ showing the relationship between spot and par forward values.

The classical way of deriving this involves considering a trading strategy in a bond and stock to replicate the forward contract. This, and the above examples, show the SUM provides a simple unified model for producing correct valuations.

Risky Zero

If the zero coupon bond $D(u)$ can default at random time T and pay fixed recovery R at that time then it has a cash flow R at T if $T \leq u$ or a cash flow 1 at u if $T > u$. Its price satisfies $D_t(u, T, R) D_t = [R 1(T \leq u) D_T + 1(T > u) D_u]|_{\mathcal{A}_t}$.

If T is exponentially distributed with $P(T > t) = e^{-\lambda t}$ and $D_t = D(t)$ is not stochastic then

$$D_0(u, T, R) = R(1 - e^{-\lambda u})D(u) + e^{-\lambda u}D(u) = [R + (1 - R)e^{-\lambda u}]D(u).$$

If $R = 1$ or $\lambda = 0$ this is just $D(u)$.

For small λ we have $D_0(u, T, R) \approx e^{(1-R)\lambda u} D(u)$ yielding a simple back-of-the-envelope approximation for the credit spread $s = (1 - R)\lambda$. If $R = 1$ or $\lambda = 0$ then $s = 0$.

This sweeps under the rug the fact we must extend our sample space to include default and recovery. We should augment the sample space by the product $[0, \infty) \times \{R\}$ and define information available at time t for the default time $T \in [0, \infty)$. A natural choice for this is the partition $\mathcal{A}_t = \{\{s\} \mid s < t\} \cup \{[t, \infty)\}$ – if default occurs prior to t we know exactly when it happened, otherwise we only know $T \in [t, \infty)$.

Even this is not a realistic model. For example, we may want to update the default rate λ as more information becomes available. A more accurate model should allow random recovery $R \in [0, 1]$ and specify joint distributions for all the random variables involved. This would completely define the value of a risky zero coupon bond, however the software implementation and fitting market data to model parameters would be challenging.

Remarks

This model ignores many salient features of how markets actually work.

Transactions have a bid/ask spread that tends to increase with the size of the trade.

Prices are not real numbers – they are integral multiples of minimum trading increment, or tick size. Likewise for trading sizes. Also, at some point no more instruments are available for trading. This is an actual problem for large hedge funds.

The definition of arbitrage as $A_{\tau_0} > 0$ and $A_t \geq 0$ thereafter is insufficient. Traders and risk managers will calculate $\|A_0\| = \|- \Gamma_0 \cdot X_0\| \leq \|\Gamma_0\| \|X_0\|$. The “arbitrage” will not be considered if the left-hand side gain is small compared the ballpark estimate of the amount of capital tied up for the initial hedge on the right-hand side.

We completely ignore the entities involved in trading. Different counterparties may have to pay different prices or additional side fees for the same instrument depending on their credit worthiness. They might even be unable to buy an instrument due to regulatory requirements. In the not too distant past banks were forbidden from purchasing stocks.

The largest lacuna, by far, in this theory is tax considerations. When you are in a 40% tax bracket adding the fifth decimal point of precision to your valuation routines is not your biggest problem.

The trajectory of mathematical finance is to develop mathematical models that can more accurately describe the realities of trading. Advances in compute power, memory, and AI are in their early stages of application. This short note provides a rigorous mathematical foundation to keep moving in that direction. For full details see the Simple Unified Model.

Appendix

We require only basic facts from finitely additive measure theory. The Banach space of bounded function on set S is

$$B(S) = \{f: S \rightarrow \mathbf{R} \mid \|f\| = \sup_{s \in S} |f(s)| < \infty\}$$

Its vector space dual can be identified with the Banach space of finitely additive measures on S , $ba(S)$. See (Dunford and Schwartz 1958) Volume I, Chapter IV.

Finitely additive measures can be multiplied by bounded functions. Define $M_g: B(S) \rightarrow B(S)$ for $g \in B(S)$ by $M_g f = fg$. Its adjoint $M_g^*: ba(S) \rightarrow ba(S)$ satisfies $\langle M_g f, \lambda \rangle = \langle f, M_g^* \lambda \rangle$. We write $g\lambda$ for $M_g^* \lambda$.

Given a positive measure having mass one the *conditional expectation* of a random variable $X \in B(S)$ given an algebra $\mathcal{A}$ is defined by $Y = E[X \mid \mathcal{A}]$ if and only if Y is $\mathcal{A}$ -measurable and $E[Y1_A] = E[X1_A]$ for all $A \in \mathcal{A}$. This is equivalent to $Y(P|_{\mathcal{A}}) = (XP)|_{\mathcal{A}}$ where the vertical bar indicates restriction. Instead of computing conditional expectation we only need restriction of measure.

Given a sample space Ω of possible outcomes and a filtration of increasing algebras $(\mathcal{A}_t)_{t \in T}$ representing information available at times $t \in T$, we assume everything is finite. Classical results can be obtained from appropriate limit arguments but we are only interested implementing the mathematics in software. Everything is finite on a computer.

If an algebra of sets is finite then its atoms form a partition. The algebra is generated by its atoms so we work directly with the partition instead of the algebra. A function is measurable with respect to an algebra if and only if it is constant on atoms. In this case it *is* a function of the atoms and we write $X: \mathcal{A} \rightarrow \mathbf{R}$.

References

- Black, Fischer, and Myron Scholes. 1973. “The Pricing of Options and Corporate Liabilities.” *Journal of Political Economy* 81 (3): 637–54. <http://www.jstor.org/stable/1831029>.
- Breeden, Douglas T., and Robert H. Litzenberger. 1978. “Prices of State-Contingent Claims Implicit in Option Prices.” *Journal of Business* 51 (4): 621–51.
- Dunford, Nelson, and Jacob T Schwartz. 1958. *Linear Operators i*. Interscience Publishers.
- Harrison, Michael, and David M. Kreps. 1979. “Martingales and Arbitrage in Multiperiod Securities Markets.” *Journal of Economic Theory* 20: 381–408. [https://doi.org/10.1016/0022-0531\(79\)90043-7](https://doi.org/10.1016/0022-0531(79)90043-7).
- Merton, Robert C. 1973. “Theory of Rational Option Pricing.” *Bell Journal of Economics and Management Science* 4 (1): 141–83. <https://www.jstor.org/stable/3003143>.
- Ross, Stephen A. 1978. “A Simple Approach to the Valuation of Risky Streams.” *The Journal of Business* 51 (3): 453–75. <https://www.jstor.org/stable/2352277>.